

Sr. No. _____

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Let us try to model cardiac arrest as a Bernoulli Discrete Hawkes Process.

In a Bernoulli B Discrete Hawkes Process (BDHP), at each discrete time step t , binary variable Y_t follows a Bernoulli conditioned on past information F_{t-1} .

$$Y_t | F_{t-1} \sim \text{Bern}(p_t)$$

We can construct p_t as -

$$p_t = \psi \left(\mu + \sum_{k=1}^{\infty} w_k Y_{t-k} \right)$$

where $\mu \rightarrow$ Baseline probability of Cardiac Arrest

$w_k \rightarrow$ excitation weight ; $w_k \geq 0$

$\psi \rightarrow$ link function mapping function into $[0, 1]$.

Over here we are going to be choosing our link function as -

$$\psi = \frac{1}{1 + e^{-x}} = \sigma(x) \rightarrow \text{Logistic Model.}$$

As we know that once a person has a Cardiac Arrest, it is highly likely that the probability of another arrest is increased and do not decay easily.

So, the excitation Kernel chosen over here must also reflect the same.

$$w(t) = \frac{\alpha}{(t + \frac{c}{\beta})^\beta}$$

where α controls the overall scale or intensity

c small constant to prevent $w(t) \rightarrow \infty$ as $t \rightarrow 0$

β determines decay rate. As $\beta \rightarrow 1$, the memory of the process is longer.